

# AI-Based Restoration of Degraded Images

SEMICON India Hackathon 2026 · KLA Problem Statement 01

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## TEAM DETAILS

**Team name:** *< fill in >*

**College / Institution:** *< fill in >*

**Contact email:** *< fill in >*

**Contact phone:** *< fill in >*

## MEMBERS & ROLES

1. *< name >* — *< role >*

2. *< name >* — *< role >*

3. *< name >* — *< role >*

4. *< name >* — *< role >*

**Repository:** <https://github.com/Deven1305/GrayScale>

# Problem Statement Addressed

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AI-Based Restoration of Degraded Images — speckle + Gaussian noise + 2× downsampling

## WHY THIS MATTERS IN SEMICONDUCTOR FABS

Metrology and inspection tools are run at their throughput limit. Shorter dwell time per site means fewer collected electrons or photons, so images arrive noisy and spatially undersampled.

Restoring signal after acquisition recovers measurable detail without slowing the tool — the same wafer throughput, but usable images. That trade sits directly on fab economics: scan slower for image quality, or restore computationally and keep the tool moving.

## WHAT THE DATA ACTUALLY IS — stated honestly

The released training set is 800 natural photographs (4 crops each = 3200 pairs), not semiconductor imagery. KLA's own materials additionally show scientific/microscopy content from a different release.

The test set is promised to be out-of-distribution. **That makes cross-origin generalisation the central technical challenge, not raw in-distribution accuracy.**

### 1 · SPECKLE NOISE

- Multiplicative, signal-dependent grain
- Pushes values beyond the true range
- Removed without blurring — blur destroys the detail we are paid to recover

### 2 · GAUSSIAN NOISE

- Additive, signal-independent
- Softens edges and fine structure
- Sharpness restored without ringing or invented texture

### 3 · 2× DOWNSAMPLING

- $256 \times 256 \rightarrow 128 \times 128$  in this release
- $512 \times 512 \rightarrow 256 \times 256$  in KLA's deck
- Lost high-frequency detail reconstructed at exactly 2× the input

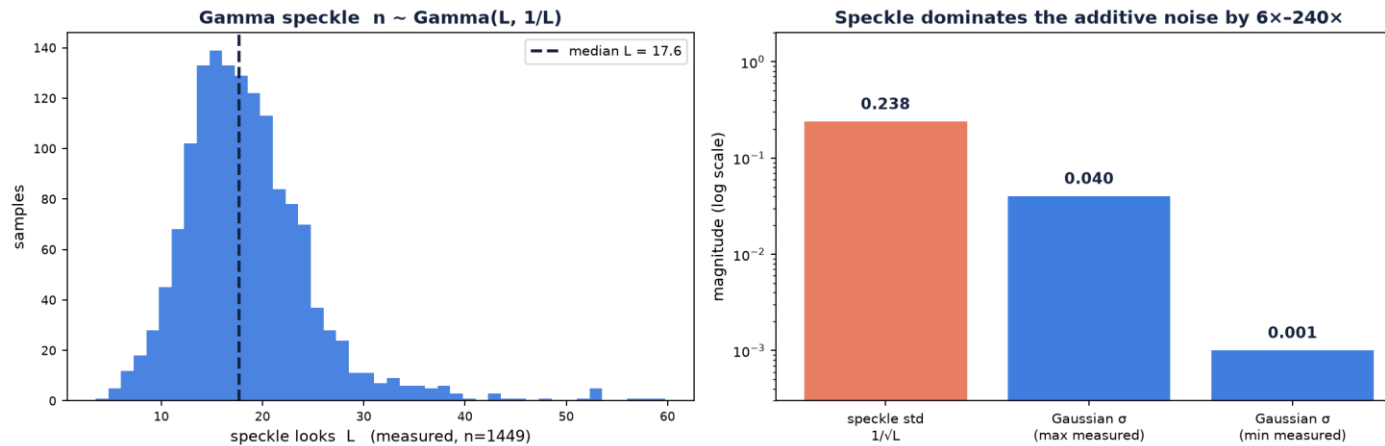
**All three may be present simultaneously on one image — but not in equal measure.** We measured speckle std  $\approx 0.24$  against additive  $\sigma$  of 0.001–0.04, so the design treats them by weight, not as three equal problems.

# Idea Description

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Build a despeckler first — the degradation is not three equal problems

**KEY FINDING:** we measured speckle std  $\approx 0.24$  against additive  $\sigma$  of 0.001–0.04 — multiplicative noise dominates by 6× to 240×.  
**Model capacity, loss design and augmentation all target it first.**



## MODEL CHOICE — NAFNet-w32

- No activation functions; LayerNorm generalises well out-of-distribution
- Best quality-per-FLOP among general restoration backbones
- Fully convolutional → both input sizes work unchanged
- SAFMN ( $\sim 0.24$  M) kept as the fast point on the Pareto curve

## WHY NOT A GAN OR DIFFUSION

- The spec forbids “artificial patterns or ringing” — exactly what adversarial training produces
- GAN training costs PSNR and SSIM, two of the three scored metrics
- Decision recorded up front rather than discovered late

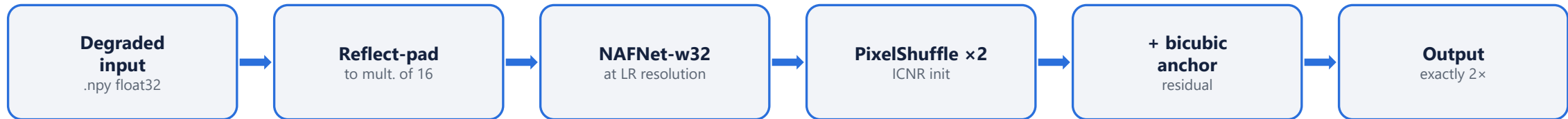
| Degradation   | How the design addresses it                                            |
|---------------|------------------------------------------------------------------------|
| Speckle       | Gamma-aware replica + log-domain unrolled prior; Charbonnier + MS-SSIM |
| Gaussian      | Same denoising path; FFT + gradient loss restore edges without ringing |
| 2× downsample | PixelShuffle(2) head on a bicubic residual anchor                      |

# Proposed Solution

Architecture · training strategy · loss design · augmentation

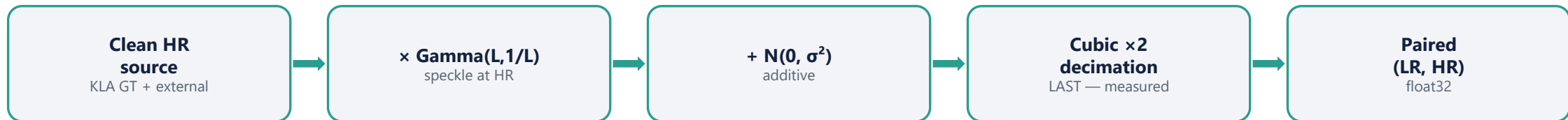
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## INFERENCE PATH



**Never clipped.** Processing stays at LR resolution throughout — upsampling first would cost  $\sim 4\times$  the compute for no quality gain.

## TRAINING DATA FLOW



**Order is fixed, not randomised** — we measured it. Order randomisation is kept as an explicit OOD augmentation switch, evaluated rather than assumed.

### TRAINING STRATEGY

- AdamW, lr 2e-4, cosine + warmup
- bf16 autocast (no GradScaler)
- EMA decay 0.999, evaluated with EMA
- Split by source\_id = index // 4

### COMPOSITE LOSS

- $1.0 \cdot \text{Charbonnier}$
- $0.2 \cdot (1 - \text{MS-SSIM})$
- $0.1 \cdot \text{FFT (L1 in frequency)}$
- $0.05 \cdot \text{gradient} + 0.01 \cdot \text{VGG}$

### AUGMENTATION

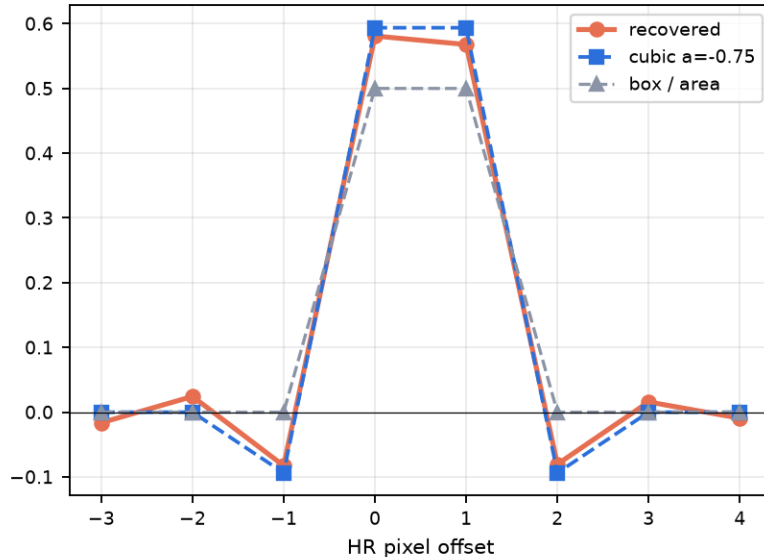
- 8-fold dihedral on paired crops
- $L \sim 8-40$ ,  $\sigma \sim \text{log-uniform}(1e-3, 4e-2)$
- Kernel  $a \sim U(-0.75, -0.45)$
- 50/50 KLA pairs vs external content

# Innovation & Uniqueness

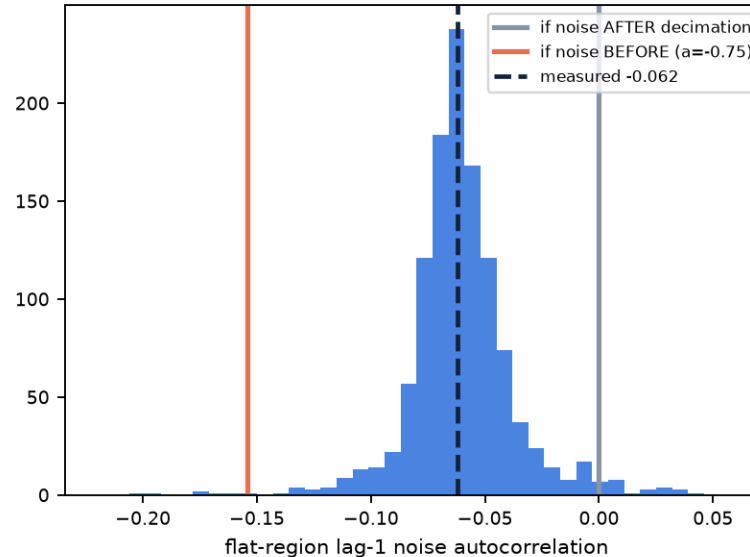
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The degradation was reverse-engineered from the data, not assumed

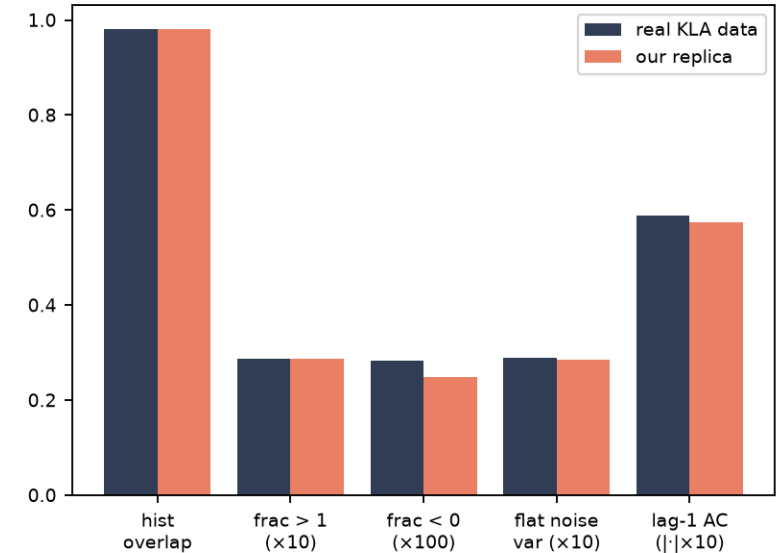
1 · Kernel recovered from  $E[y|x]$



2 · Order test — unimodal, so not randomised



3 · Replica matches real statistics



## 1 · RECONSTRUCTED DEGRADATION

- No metadata ships with the dataset — we recovered the model from the pairs
- Kernel regressed directly from  $E[y|x]$ ; box, area, Lanczos and strided decimation all excluded
- Replica hits 0.982 histogram overlap with the real data

## 2 · LOG-DOMAIN UNROLLED NET

- Built from KLA's own appendix reference (Monga et al. 2021) — a slide never presented
- $\log()$  turns multiplicative speckle additive, so HQS/ADMM machinery applies
- K analytic data-fidelity steps alternating with a learnt CNN prior

## 3 · SYNTHETIC OOD VALIDATION

- KLA withheld test ground truth — we manufacture it
- Validated replica applied to external corpora gives labelled OOD families
- Degradation-consistency check scores unlabelled test images with no GT

Every number measured on held-out data. Nothing estimated.

RESTORATION QUALITY · 320 val images, held out BY SOURCE

| Method             | PSNR ↑ | SSIM ↑ | LPIPS ↓ |
|--------------------|--------|--------|---------|
| Bicubic ×2 (floor) | 23.067 | 0.5129 | 0.4425  |
| BM3D + bicubic ×2  | 25.956 | 0.6527 | 0.5576  |
| NAFNet-w48 (ours)  | 26.415 | 0.7333 | 0.2455  |

Beats every baseline on all three scored metrics: +3.35 dB / +0.220 SSIM / −0.197 LPIPS over bicubic, and +0.46 dB / +0.081 SSIM / −0.312 LPIPS over BM3D.

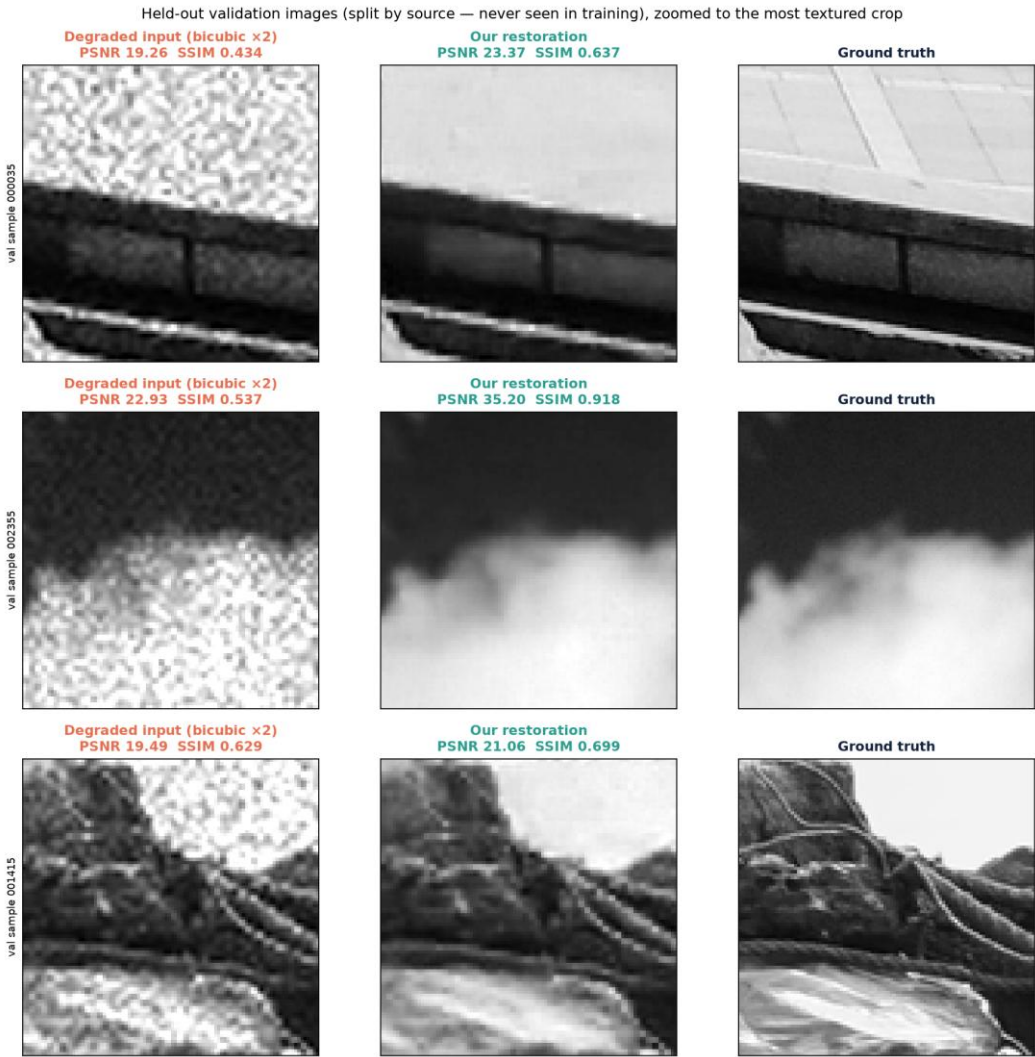
Note BM3D's LPIPS (0.558) is worse than plain bicubic (0.443): it buys PSNR by over-smoothing, which PSNR rewards and LPIPS punishes — the failure the spec warns against, quantified.

OUT-OF-DISTRIBUTION · synthetic GT from our own replica

| Family               | PSNR ↑ | SSIM ↑ | LPIPS ↓ |
|----------------------|--------|--------|---------|
| Urban100 (buildings) | 24.432 | 0.7750 | 0.2204  |
| BSD100 (natural)     | 26.422 | 0.7506 | 0.2494  |
| Set14                | 26.655 | 0.7643 | 0.2110  |

Urban100 drops 1.98 dB — a genuine OOD penalty on the buildings case KLA named. BSD100 and Set14 hold. Ground truth exists only because we reconstructed the degradation.

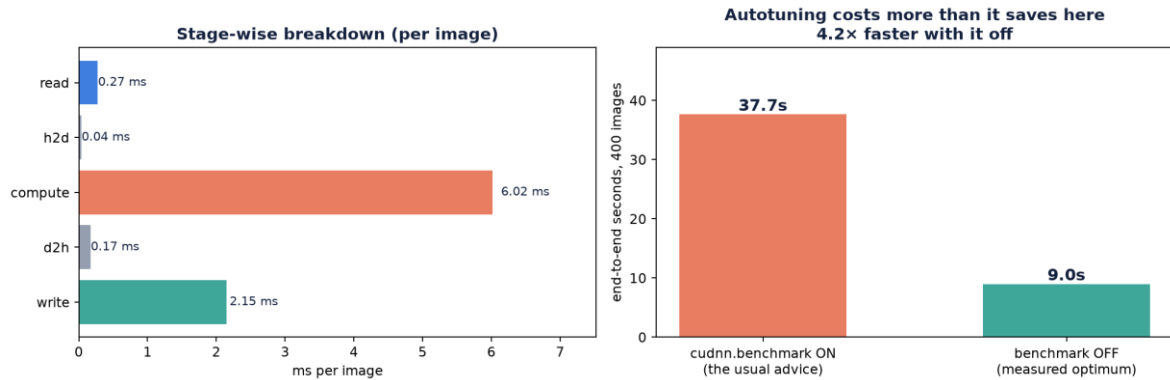
Degradation replica fidelity: 0.982 histogram overlap · frac>1 0.0287 vs 0.0288 · frac<0 0.00249 vs 0.00282 · noise lag-1 −0.0574 vs −0.0588



Held-out validation crops — never seen in training.

TECH STACK

| Component        | Choice                                     |
|------------------|--------------------------------------------|
| Framework        | PyTorch 2.11.0 + CUDA 12.8                 |
| Training GPU     | consumer GPU, 8 GB (recent architecture)   |
| Benchmark target | NVIDIA H100 (KLA-side)                     |
| Precision        | bf16 autocast (train) · fp16 (inference)   |
| Environment      | conda, Python 3.11.15, 115 pinned packages |
| Metrics          | torchmetrics · lpips · pyiqa · bm3d        |



MEASURED PERFORMANCE · 400-image test set

| Measure                 | Value                            |
|-------------------------|----------------------------------|
| Model parameters        | 15.24 M                          |
| Weight file (fp16)      | 30.6 MB (8× smaller than ckpt)   |
| Training time           | 40 epochs, ~1.6 h on an 8 GB GPU |
| Peak training VRAM      | 7.1 GB of 8 GB                   |
| Inference, per image    | 22.5 ms end-to-end               |
| Full test set (400 img) | 9.0 s · 44.4 img/s               |

WHAT THE MEASUREMENT CHANGED

Two defaults measured, then reversed.

cudnn autotuning costs 11.7 s of warmup for an ~8 % main-loop gain — a 2.4x end-to-end LOSS at this scale. And DataLoader workers are slower where they spawn, because each re-imports torch: 35.2 s with 4 workers vs 16.4 s with none.

**Both are now platform-aware flags, not hardcoded**, since a Linux H100 forks workers cheaply and may flip the answer back.

## GITHUB REPOSITORY (mandatory)

<https://github.com/Deven1305/GrayScale>

**Visibility must be set to PUBLIC before submission** — a private repository cannot be benchmarked, and unscored submissions cannot win.

## DEMONSTRATION VIDEO (optional, recommended)

◀ *TO BE ADDED* ▶

## REQUIRED REPOSITORY CONTENTS

| # | Item                                          | Status    |
|---|-----------------------------------------------|-----------|
| 1 | README.md — clone-and-run instructions        | delivered |
| 2 | Evaluation script (standalone .py, dir → dir) | delivered |
| 3 | Training script (train.py, config-driven)     | delivered |
| 4 | Trained model weights (10 MB fp16)            | delivered |
| 5 | Restored test outputs (400 images)            | delivered |
| 6 | requirements.txt (pip freeze)                 | delivered |



## CITED BY KLA IN THE PROBLEM-STATEMENT MATERIALS

- [1] T. Kumar, R. Brennan, A. Mileo, M. Bendeckache. "Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions." IEEE Access, vol. 12, 2024.
- [2] L. Zhai, Y. Wang, S. Cui, Y. Zhou. "A Comprehensive Review of Deep Learning-Based Real-World Image Restoration." IEEE Access, vol. 11, pp. 21049–21067, 2023.
- [3] J. Terven, D. M. Cordova-Esparza, J. A. Romero-González et al. "A Comprehensive Survey of Loss Functions and Metrics in Deep Learning." Artificial Intelligence Review 58, 195, 2025.
- [4] V. Monga, Y. Li, Y. C. Eldar. "Algorithm Unrolling: Interpretable, Efficient Deep Learning for Signal and Image Processing." IEEE Signal Processing Magazine, vol. 38, no. 2, pp. 18–44, 2021.

## METHODS

- [5] L. Chen, X. Chu, X. Zhang, J. Sun. "Simple Baselines for Image Restoration." ECCV 2022. (NAFNet)
- [6] K. Zhang, J. Liang, L. Van Gool, R. Timofte. "Designing a Practical Degradation Model for Deep Blind Image Super-Resolution." ICCV 2021. (BSRGAN)
- [7] K. Zhang, L. Van Gool, R. Timofte. "Deep Unfolding Network for Image Super-Resolution." CVPR 2020. (USRNet)
- [8] L. Sun, J. Dong, J. Tang, J. Pan. "Spatially-Adaptive Feature Modulation for Efficient Image Super-Resolution." ICCV 2023. (SAFMN)

## DATASETS & TOOLS

- [9] E. Agustsson, R. Timofte. "NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study." CVPRW 2017. (DIV2K)
- [10] J.-B. Huang, A. Singh, N. Ahuja. "Single Image Super-Resolution from Transformed Self-Exemplars." CVPR 2015. (Urban100)
- [11] M. Cimpoi et al. "Describing Textures in the Wild." CVPR 2014. (DTD)
- [12] K. Dabov et al. "Image Denoising by Sparse 3-D Transform-Domain Collaborative Filtering." IEEE TIP, 2007. (BM3D)
- [13] R. Zhang et al. "The Unreasonable Effectiveness of Deep Features as a Perceptual Metric." CVPR 2018. (LPIPS)

### OUR OWN ANALYSIS

- docs/forensics\_report.md — degradation model reconstructed from the paired data
- 25 standalone, re-runnable forensics scripts
- All figures on this deck generated from measured data